

RAHUL MITRA

OPERATIONS RESEARCH ENGINEER PROFILE

Singapore | mitrarahul2002@gmail.com | www.linkedin.com/in/rahulmitra-dev | github.com/Rah-Rah-Mitra | rahul-mitra.vercel.app

PROFILE SUMMARY

Early-career operations research engineer combining NUS ISE foundations with deployable software, simulation, optimization, and analytics automation for scheduling, digital twin, process improvement, and decision-support roles.

EDUCATION

National University of Singapore

Aug 2023 - Jul 2027

- Bachelor of Engineering, Industrial Systems Engineering; Second Major in Computer Science; Minor in Mathematics
- Relevant coursework: Operations Research; Simulation; Statistics for Engineering Applications; Industrial Systems Engineering; Artificial Intelligence; Software Engineering

Anderson Serangoon Junior College

Feb 2019 - Dec 2020

- Singapore-Cambridge GCE A-Level

WORK AND INTERNSHIP EXPERIENCE

Data Analytics and Product Management Intern, Abbott Laboratories Singapore Pte Ltd

Jan 2026 - Present

- Built a Python digital twin simulation and constraint-programming optimizer for an NP-hard hybrid flow shop scheduling problem, translating production complexity into decision-ready schedules.
- Architected an Azure-based web simulator for Advanced Process Control systems across global manufacturing plants, using Docker containerization and Bitbucket CI/CD for repeatable releases.
- Offered a follow-on 7-month Data Engineer, Operational AI Systems contract, extending work across manufacturing analytics, deployment reliability, and operational AI workflows.

Base Support Assistant, Republic of Singapore Navy

Jan 2022 - Jan 2023

- Engineered an Excel VBA automation workflow that compressed a six-month food-waste analysis cycle into a sub-three-minute reporting process inside a security-constrained environment.
- Managed sensitive operational data under SAF confidentiality expectations while improving reliability, traceability, and analyst capacity.

Software Engineer (Citizen Developer), People's Association

Sep 2025 - Present

- Awarded the S\$20,000 Sparks Community Innovation Fund at the 2025 Community Hackathon to continue Churp for Tengah community gardening operations.
- Built product workflows for transparent plot allocation, resident feedback, and civic service delivery while conducting user testing to refine product requirements.

SELECTED PROJECTS AND COMPETITIONS

Hybrid Flow Shop Digital Twin Optimizer

2026

- Modeled production scheduling as a digital twin simulation and constraint-programming optimization problem, connecting ISE methods with deployment-oriented Python tooling.

Volt Pulse SG - SMU Hack For Cities Finalist

Jan 2026

- Reached Top 8 of 50+ teams with an agentic AI platform for HDB energy-cost tracking using SEALION embeddings, Supabase vector search, RRF routing, and scheduler-agent planning.

IE2110 Operations Research Graph Optimization

2024

- Applied IE2110 Operations Research I concepts from linear programming, network flow models, and nonlinear programming; implemented Jupyter SSSP/APSP graph optimization for engineering decision support.

NVIDIA Disaster Risk Monitoring with Satellite Imagery

2023

- Completed hands-on geospatial AI training covering satellite imagery, U-Net semantic segmentation, disaster-risk assessment, and GPU-accelerated model workflows.

TECHNICAL SKILLS AND CERTIFICATIONS

Skills: Constraint Programming; Scheduling Optimization; Hybrid Flow Shop; Digital Twin Simulation; Dijkstra; SSSP/APSP; Python; Jupyter; Statistics; Simulation; Excel VBA; Process Improvement; Decision Analytics.

- Excel 2019 Advanced - analytics automation and spreadsheet modeling.
- NVIDIA Disaster Risk Monitoring Using Satellite Imagery - geospatial AI, U-Net, satellite data, GPU workflows.
- Packt Deep Reinforcement Learning Hands-On - DQN, Double DQN, DDPG, PPO, A2C, Python implementation.

CO-CURRICULAR AND COMMUNITY

Elderly Care Volunteer, NTUC Health Active Ageing Centre

Sep 2025 - Present

- Supported senior engagement activities, strengthening empathy for healthcare, accessibility, and community product design.

Astronomy Head and Robotics Member, STEM Inc. (ASRJC)

2019 - 2020

- Led astronomy outreach, competed in the Singapore Astronomical Olympiad, and built autonomous robotics systems for national competitions.